

MATHEMATICAL DESIGN AND FORMAL VERIFICATION OF SECURE MESSAGING AND INTEGRITY SYSTEM

- Module 3 – Combinatorics & Counting
- Mathematics for Computation

BATCH NO:2

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PROBLEM STATEMENT

- A secure messaging and integrity system contains passwords, cryptographic keys, message configurations and hash values with many possible combinations.
- There is a need to mathematically count these configurations and analyse collision situations.
- This project applies counting principles, permutations, combinations, inclusion–exclusion, the generalized Pigeonhole Principle and birthday-problem counting to study key-space size and collision-related situations.

OBJECTIVES

- Apply combinatorial principles to security-related configurations.
- Calculate possible password and cryptographic key combinations.
- Analyse arrangements and selections using permutations and combinations.
- Use inclusion–exclusion and the Pigeonhole Principle to study collisions.
- Understand key-space sizing and collision counting mathematically.

FUNDAMENTAL COUNTING PRINCIPLES

- Sum Rule: total ways = $m + n$ when choices are mutually exclusive.
- Product Rule: total ways = $m \times n$ for independent stages.
- Counting by cases divides a complex problem into separate cases.
- Application: counting possible security-code and key configurations.

PERMUTATIONS AND COMBINATIONS

- Permutation: arrangement where order matters. $nPr = n!/(n-r)!$
- Combination: selection where order does not matter. $nCr = n!/[r!(n-r)!]$
- These methods count possible ordered and unordered security configurations.

REPETITION AND CIRCULAR ARRANGEMENTS

- Permutations with repetition: $n!/(n_1!n_2!\dots n_k!)$.
- Circular permutations: $(n - 1)!$.
- Combinations with repetition: $C(n+r-1, r)$.
- These techniques handle repeated and circular configurations.

STARS AND BARS

- Stars-and-Bars counts distributions of identical objects among groups.
- Formula: $C(n+r-1, r-1)$.
- It can model structured distribution problems in computing and security-related configurations.

BINOMIAL THEOREM AND PASCAL'S IDENTITY

- Binomial Theorem: $(x+y)^n = \sum C(n,r)x^{n-r}y^r$.
- Pascal's Identity: $C(n,r) = C(n-1,r-1) + C(n-1,r)$.
- Both connect algebraic expressions with combinatorial counting.

INCLUSION–EXCLUSION AND DERANGEMENTS

- Inclusion–Exclusion prevents double counting.
- $|A \cup B| = |A| + |B| - |A \cap B|$.
- Derangements are arrangements in which no element remains in its original position.
- These methods help count constrained configurations.

GENERALIZED PIGEONHOLE PRINCIPLE

- If more than $n \times k$ objects are placed into n boxes, at least one box contains more than k objects.
- It establishes when repeated assignments or collisions are unavoidable.
- This provides a mathematical basis for collision analysis.

KEY-SPACE SIZING

- Key-space is the total number of possible keys or security configurations.
- For n possible symbols at each of L positions: total keys = n^L .
- Example: 4-digit code using 10 digits $\rightarrow 10^4 = 10,000$ possibilities.
- Counting provides a mathematical measure of configuration-space size.

BIRTHDAY PROBLEM AND COLLISION COUNTING

- The birthday problem studies repeated values among many choices.
- Similar counting reasoning helps analyse hash collisions.
- When the input space is larger than the output space, collisions are unavoidable by the Pigeonhole Principle.
- Module 3 uses this as combinatorial collision reasoning.

RECURRENCE, GENERATING FUNCTIONS & CATALAN NUMBERS

- Recurrence relations define counting quantities using earlier values.
- Generating functions represent sequences algebraically:
 $A(x) = a_0 + a_1x + a_2x^2 + \dots$
- Catalan numbers: $C_n = \frac{1}{n+1} \times C(2n, n)$.
- These are additional tools for structured counting problems.

APPLICATION TO SECURE MESSAGING

- Count possible passwords, keys and security configurations.
- Analyse possible hash collisions using counting principles.
- Use the Pigeonhole Principle to identify unavoidable collisions.
- Use combinatorial methods to mathematically analyse the size of security spaces.

CONCLUSION

- Combinatorics provides a mathematical foundation for analysing secure messaging and integrity systems.
- Counting principles determine the number and structure of possible security configurations.
- Key-space sizing and collision counting connect Module 3 directly to cryptographic security analysis.
- The project demonstrates how mathematical counting supports the design and formal analysis of security mechanisms.